The Financial Impacts of Decision-Making Frictions

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October 2026

Preliminary and Incomplete ☺

Motivation

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

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2. Which frictions have the **largest** impacts?
3. How do the impact of these frictions **vary** across individuals?

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- **Challenge**: frictions affect many different outcomes, and no common **metric**
  - Example: how to compare changes in labor income, returns, and life expectancy?

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  - Example: how to compare changes in labor income, returns, and life expectancy?
- **This paper**: propose and measure one metric across frictions = financial impact

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Takeaway: effects of frictions are sizeable, regressive, and heterogeneous

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Our Definition of a Friction

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Scope: estimate the financial impact of a large set of frictions, **not**

- Evaluate specific policies (Hendren-Sprung-Keyser 20)
- Measure the effects of specific biases (Stango-Zinman 23)
- Assess the welfare implications of removing frictions (Bernheim-Taubinsky 18)

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3. If choice $\neq$ friction, estimates are lower bound on utility gains to rationalize it

List of Frictions That We Consider

No.	Category	Friction	No.	Category	Friction
1	Borrowing	Paying Late Fees	11	Savings	Investing in High-Fee Mutual Funds
2	Borrowing	Paying Overdraft Fees	12	Insurance	Choosing Dominated Health Insurance
3	Borrowing	Suboptimally Repaying Credit Card Debt	13	Insurance	Suboptimally Lapsing Life Insurance
4	Borrowing	Not Refinancing a Mortgage	14	Benefits	Not Claiming EITC Benefits
5	Borrowing	Not Getting an Additional Mortgage Quote	15	Benefits	Not Claiming SNAP Benefits
6	Borrowing	Not Filing for Bankruptcy	16	Education	Not Attending College Due to Frictions
7	Savings	Not Meeting the Employer 401(k) Match	17	Education	Attending 2-Year For-Profit Colleges
8	Savings	Not Investing in Stock Mutual Funds	18	Health	Not Taking Statin Medications
9	Savings	Saving in Low Interest Accounts	19	Health	Not Screening for Colon Cancer
10	Savings	Trading Individual Stocks	20	Health	Failing to Quit Smoking

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1. For some choices, we only consider a **subset** of them to be **frictions**
   - **Not Attending College**: only measure non-attendance due to information frictions
   - **Not Filing for Bankruptcy**: only consider non-filing due to incorrect beliefs
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2. Notable examples of choices that we **do not** classify as frictions
   - **Gambling**: normative utility may include anticipatory or entertainment utility
   - **Using Payday Loans**: may be optimal if sufficiently credit-constrained

Outline

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2. Measurement and Examples
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Environment

- **Individual** i observed at age t in group j_{it} with survival $A_{it} \equiv \mathbf{1}$ (i alive at t)
- $s_{it} \equiv \mathbf{1}$ (i at risk of friction at t), $f_{it} \equiv \mathbf{1}$ (i subject to friction at t)
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  - $R_{is}^{(t)}$ and $A_{is}^{(t)}$ = counterfactual values at $s \in [t, t + H - 1]$ ($H = 1$ for most frictions)

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Welfare Gain from Removing a Particular Friction

Assume that:

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General Approach

Steps for Measuring the Financial Impact of Each Friction

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Groups = {bottom 25%, middle 50%, top 25%} age-specific pre-tax/transfer income

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Not Investing in Stock Mutual Funds: reduces Capital Income

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Not Claiming EITC Benefits: reduces Benefits

Definition of Individual Financial Impact

$$\ell_{it} \equiv \sum_{h=0}^{H-1} \left[A^{(t)}_{i,t+h} R^{(t)}_{i,t+h} - A_{i,t+h} R_{i,t+h} \right]$$

1. Consumption and wealth not observed, so construct consumable income directly:

$$R_{it} = \text{Labor Income}_{it} + \text{Capital Income}_{it} + \text{Benefits}_{it} - \text{Taxes}_{it} - \lambda_{it}$$

2. Compute mechanical effects of different frictions through different channels

Paying Credit Card Late Fees: creates direct losses from additional fees

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2. Compute mechanical effects of different frictions through different channels

Failing to Quit Smoking: creates direct losses from cigarettes and increases mortality

Mechanical Effects We Include Across Frictions

No.	Category	Friction	Source of Individual Financial Impact: Labor Income$_{it}$	Capital Income$_{it}$	Benefits$_{it}$	Taxes$_{it}$	Direct Losses λ_{it}	Mortality A_{it}	Dynamic Effects $H > 1$
1	Borrowing	Paying Late Fees					✓		
2	Borrowing	Paying Overdraft Fees					✓		
3	Borrowing	Suboptimally Repaying Credit Card Debt					✓		
4	Borrowing	Not Refinancing a Mortgage					✓		✓
5	Borrowing	Not Getting an Additional Mortgage Quote					✓		✓
6	Borrowing	Not Filing for Bankruptcy Due to Frictions					✓		
7	Savings	Not Meeting the Employer 401(k) Match	✓						
8	Savings	Not Investing in Stock Mutual Funds		✓		✓			
9	Savings	Saving in Low Interest Accounts		✓		✓			
10	Savings	Trading Individual Stocks		✓		✓			
11	Savings	Investing in High-Fee Mutual Funds		✓		✓			
12	Insurance	Choosing Dominated Health Insurance					✓		✓
13	Insurance	Suboptimally Lapsing Life Insurance					✓		✓
14	Benefits	Not Claiming EITC Benefits			✓				
15	Benefits	Not Claiming SNAP Benefits			✓				
16	Education	Not Attending College Due to Frictions	✓			✓	✓†		
17	Education	Attending 2-Year For-Profit Colleges	✓			✓	✓		
18	Health	Not Taking Statin Medications					✓†	✓	✓
19	Health	Not Screening for Colon Cancer					✓†	✓	✓
20	Health	Failing to Quit Smoking					✓	✓	✓

†: $\lambda_{it} < 0$; removing the friction creates direct costs rather than gains.

Consumable Income across Groups: PSID 2011-2019

$$R_{it} = \text{Labor Income}_{it} + \text{Capital Income}_{it} + \text{Benefits}_{it} - \text{Taxes}_{it} - \lambda_{it}$$

	Income Group j		
% of R_{it}	Low	Middle	High
Labor Income	40%	77%	90%
Capital Income	2%	3%	13%
Social Security + DB Pensions	37%	23%	11%
Other Benefits	22%	4%	2%
401(k)/IRA Accrual	5%	8%	12%
Taxes	−4%	−15%	−28%
Direct Losses (λ_{it})	1.4%	1.1%	0.5%

Consumable Income across Groups: PSID 2011-2019

$$R_{it} = \text{Labor Income}_{it} + \text{Capital Income}_{it} + \text{Benefits}_{it} - \text{Taxes}_{it} - \lambda_{it}$$

$$\overline{R}_j = \text{average } \textbf{permanent } R_{it} \text{ within income group } j \qquad \text{(same for all frictions)}$$

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Income Group j	Average R_{it}	$\overline{R}_j$
Low	\$19.1K	\$23.1K
Middle	\$38.3K	\$40.2K
High	\$89.7K	\$83.2K
High / Low	4.7×	3.6×

Specific Examples

Example: Paying Late Fees

- s_{it} = having a credit card, f_{it} = paying at least one late fee in the past year
  - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
  - Adjust for survey underreporting of credit cards (Zinman 2009)

- Financial impact comes from λ_{it} = direct losses from paying late fees
  - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
  - L_j = (average late fee) $\times$ (# fees per card | fees > 1) $\times$ (# cards per cardholder)

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Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.466	0.309	\$409	\$58
	(0.013)	(0.018)	[384, 429]	[49, 64]
Middle Income	0.738	0.248	\$409	\$73
	(0.004)	(0.010)	[384, 429]	[70, 78]
High Income	0.829	0.132	\$409	\$44
	(0.008)	(0.011)	[384, 429]	[38, 51]

Example: Attending 2-Year For-Profit Colleges

- s_{it} = attending a 2-year college, f_{it} = attending 2-year for-profit
  - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)

- Financial impact combines literature (Cellini-Turner 2019) + PSID + simulation
  - 1.5pp reduction in employment probability $\Rightarrow \downarrow$ Labor Income
  - 11.3 log-point earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
  - \$5,965 additional debt (repaid under default plan) $\Rightarrow \lambda_{it}$

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Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.263	0.236	\$1,813	\$113
	(0.017)	(0.015)	[1,522, 2,107]	[85, 141]
Middle Income	0.277	0.094	\$4,320	\$113
	(0.009)	(0.007)	[3,593, 5,065]	[85, 139]
High Income	0.138	0.037	\$9,165	\$46
	(0.011)	(0.013)	[7,619, 10,783]	[14, 78]

Example: Not Investing in Stock Mutual Funds

- s_{it} = financial wealth $>$ max$\{2$ mo. pre-tax salary, $\$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
  - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ Capital Income
  - Use participants' mean equity share by income group and 5-year age bin
  - Earn 5% equity premium net of fund fees and return loss on checking/savings
  - Compute ℓ_{it} as after-tax return gain multiplied by wealth

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Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.328	0.648	\$326	\$69
	(0.009)	(0.013)	[277, 404]	[57, 87]
Middle Income	0.645	0.310	\$895	\$179
	(0.006)	(0.008)	[803, 983]	[159, 200]
High Income	0.869	0.131	\$2,631	\$301
	(0.006)	(0.007)	[2,116, 3,070]	[236, 357]

Example: Failing to Quit Smoking

- s_i quit by 50 or still smoke and want to quit; $s_i f_i$ = still smoke and want to quit (NHIS)
- Choose age 50 as critical year for two reasons: (Doll et al. 2004; Pirie et al. 2013; Jha 2020)
  1. Those who successfully quit before 50 have similar mortality to never-smokers
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Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.437 (0.005)	0.624 (0.008)	$3,066 [$2,920, $3,516]	$837 [$797, $960]
Middle Income	0.452 (0.004)	0.465 (0.006)	$3,718 [$3,439, $4,317]	$781 [$723, $907]
High Income	0.396 (0.006)	0.352 (0.011)	$5,893 [$4,825, $11,680]	$822 [$673, $1,630]

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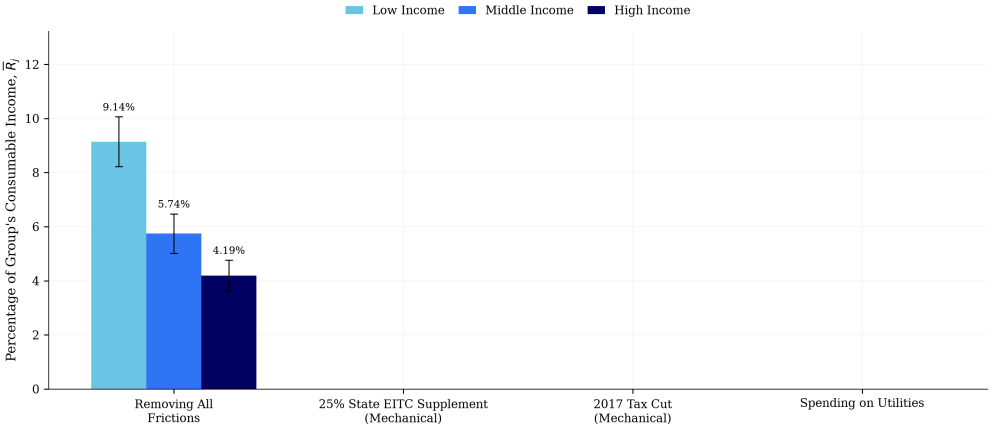
- Annualize lifetime losses by expected adult years alive in the status quo

$$L_{\text{Low}} = \underbrace{\$2{,}698}_{\text{mortality component}} + \underbrace{\$368}_{\text{cigarette spending}} = \$3{,}066 \text{ / year}$$

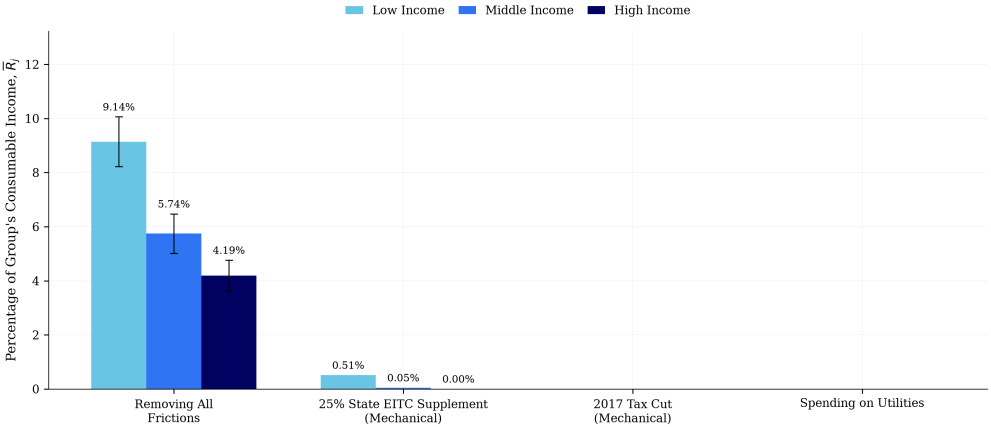
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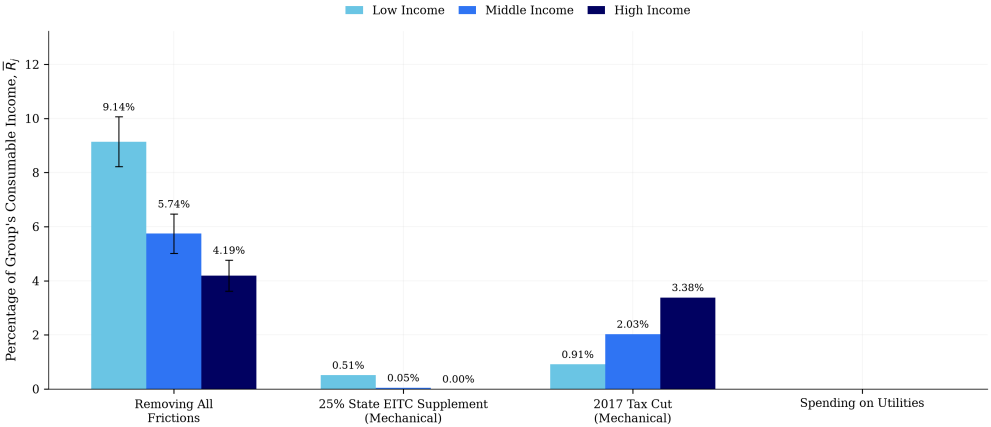
Aggregate Financial Impacts



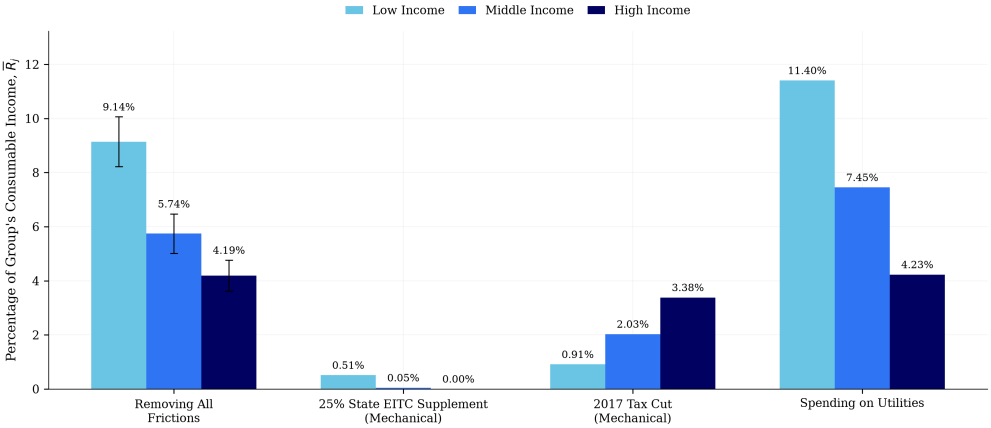
Aggregate Financial Impacts and Benchmarks



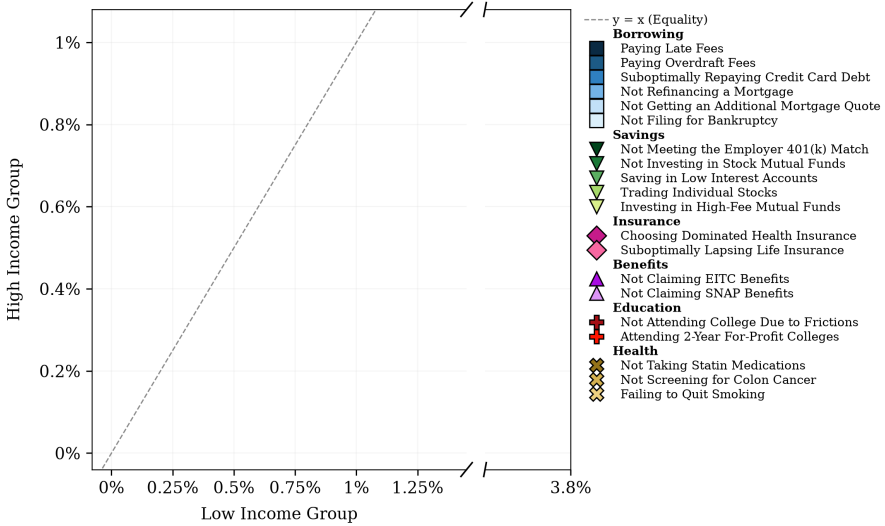
Aggregate Financial Impacts and Benchmarks



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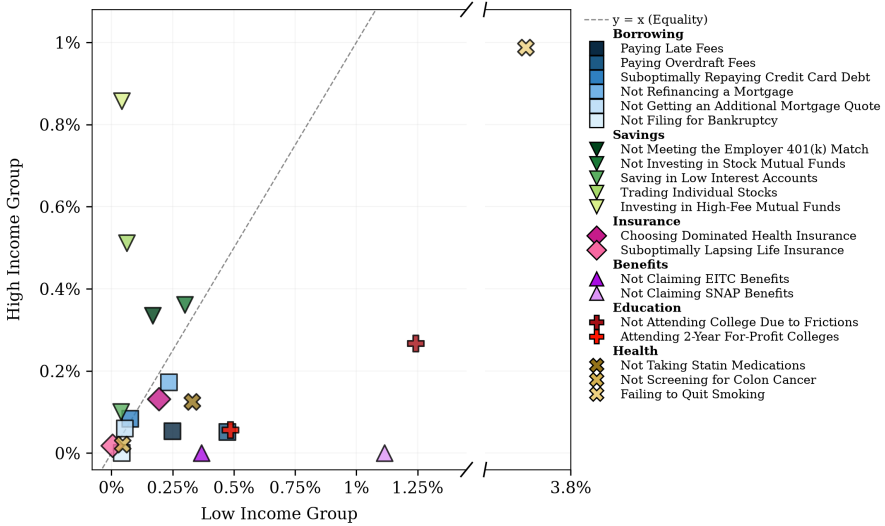


Financial Impacts Across Frictions



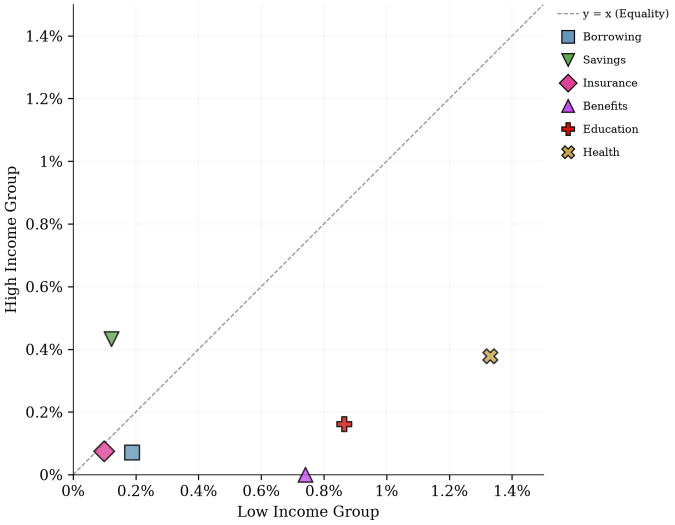
▸ Individual Estimates ▸ Dollars

Financial Impacts Across Frictions



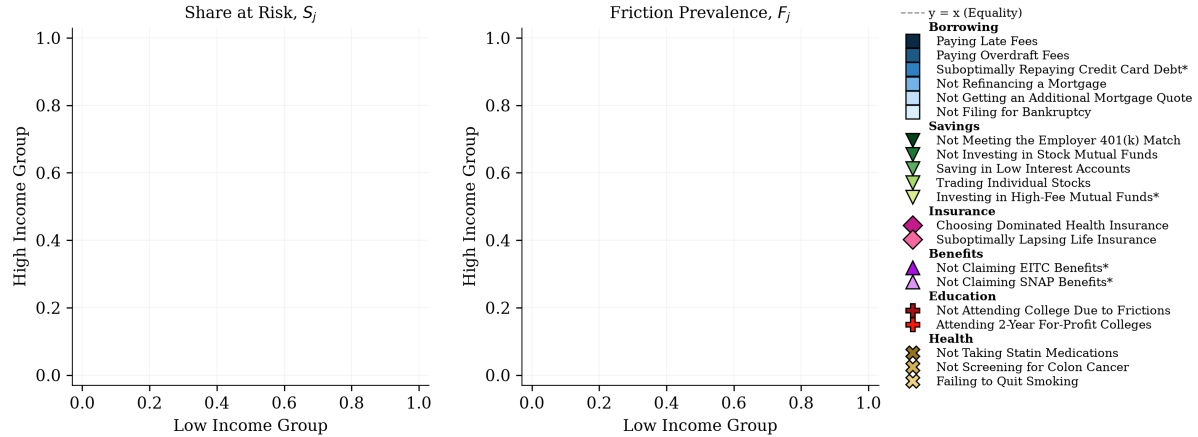
▸ Individual Estimates ▸ Dollars

Financial Impacts Across Categories

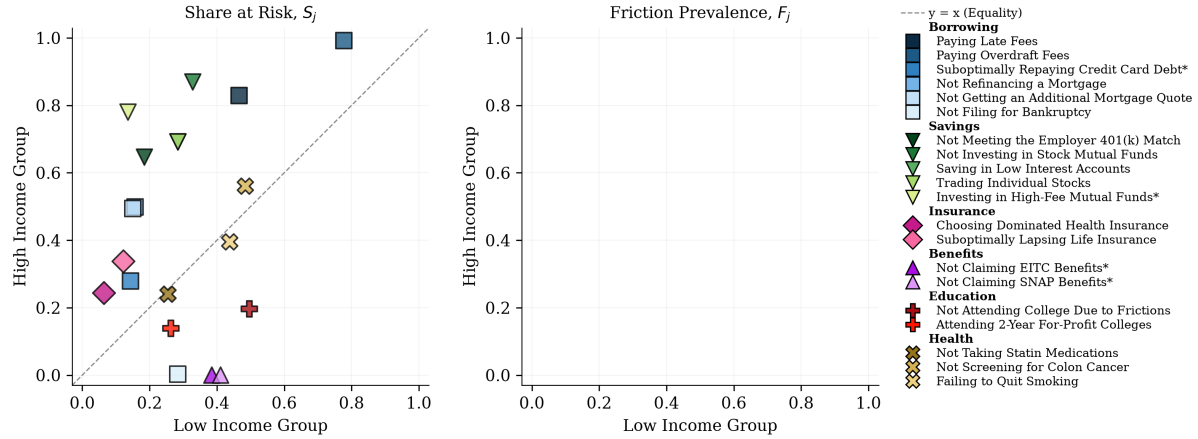


▸ Dollars ▸ Remove Categories

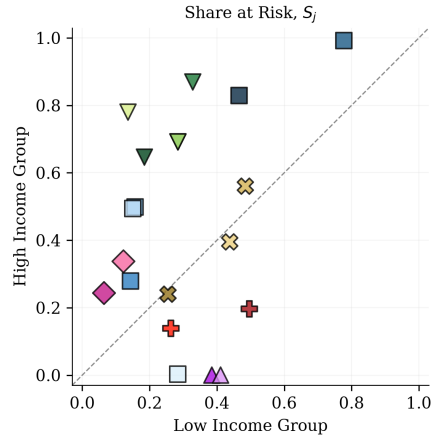
Decomposing Financial Impact: Shares and Prevalence

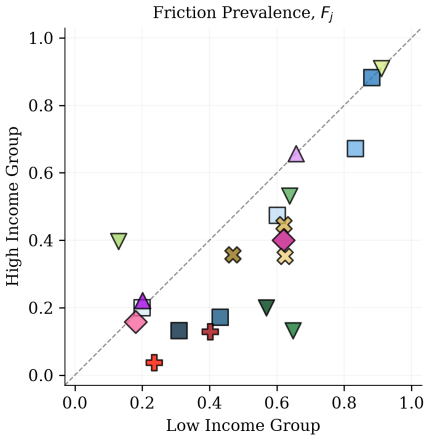


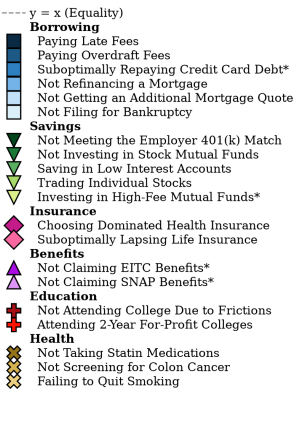
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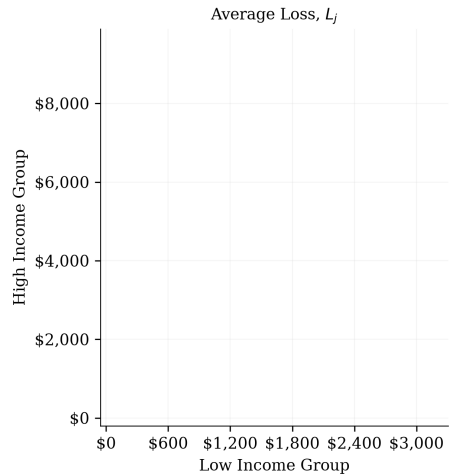
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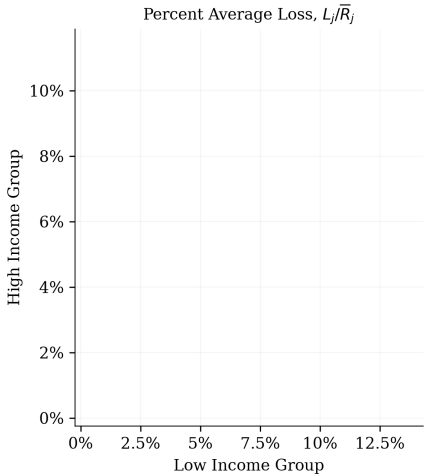


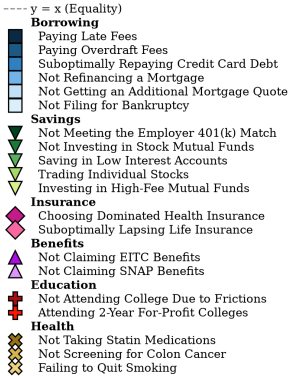




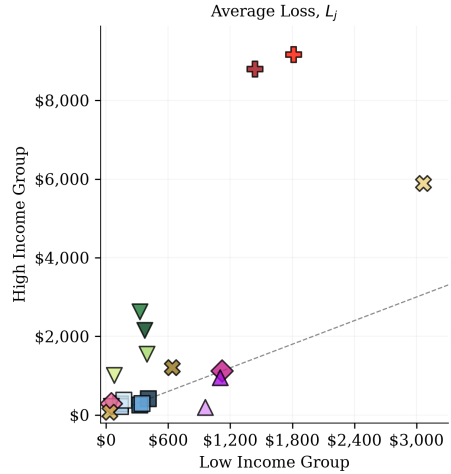
Decomposing Financial Impact: Conditional Losses

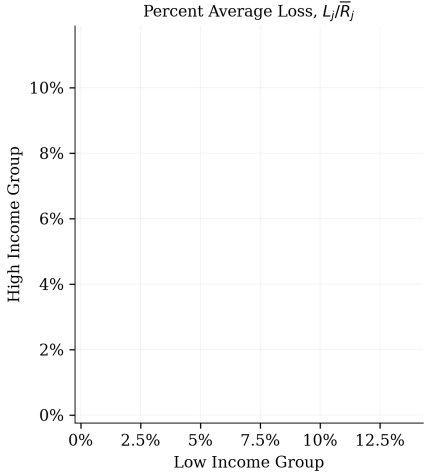


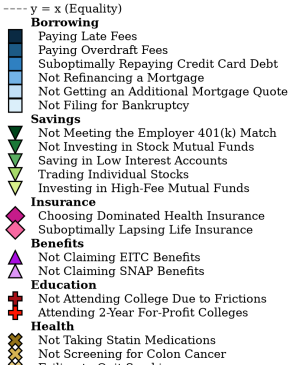




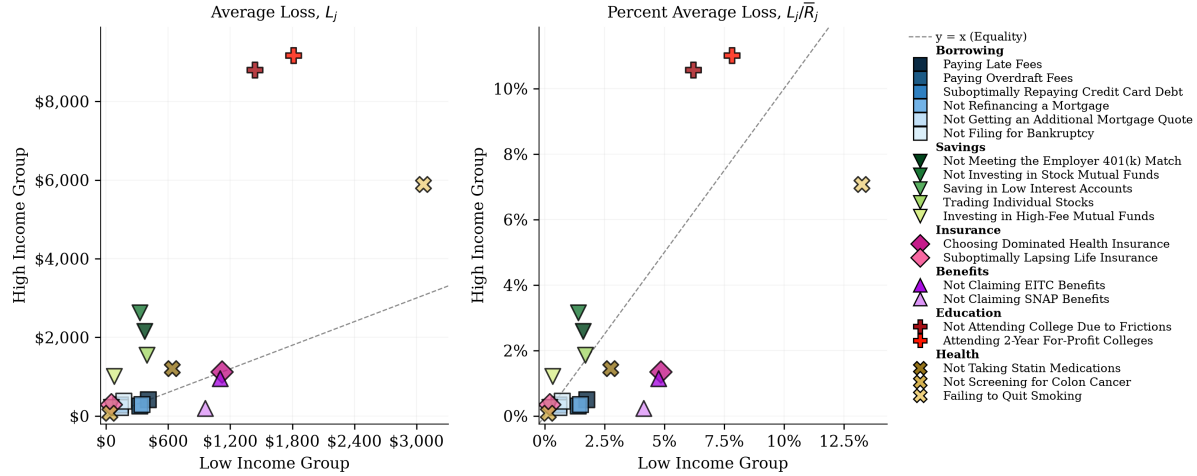
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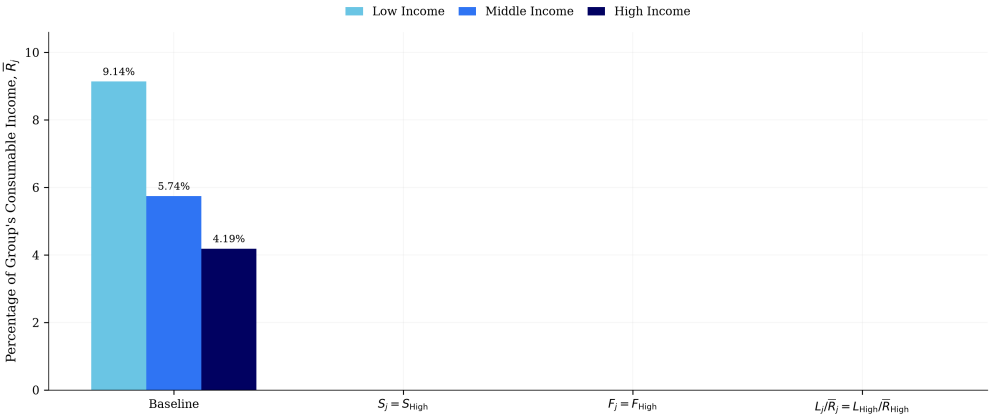




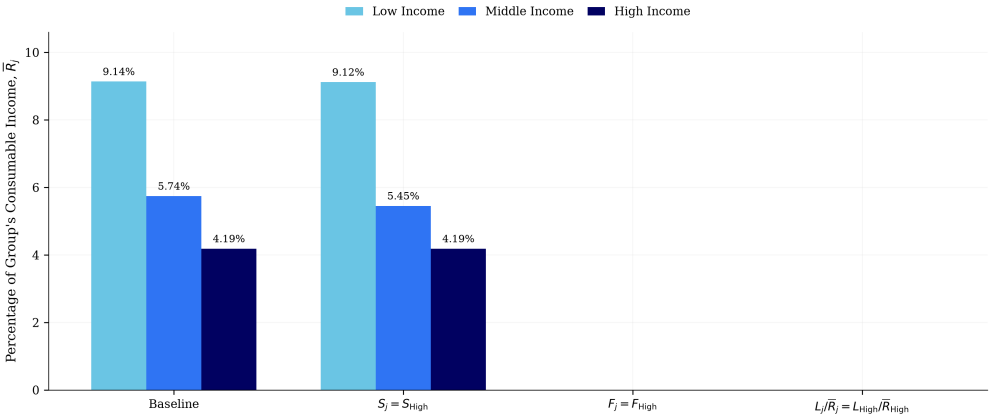
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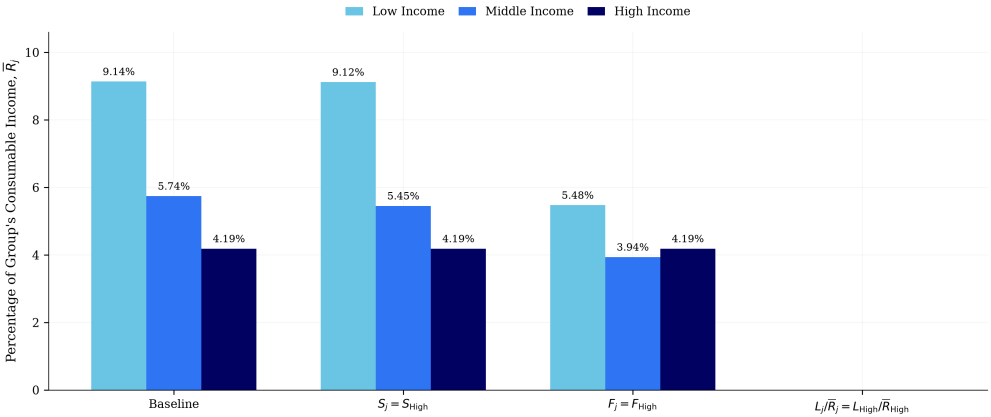
Decomposing Aggregate Financial Impacts



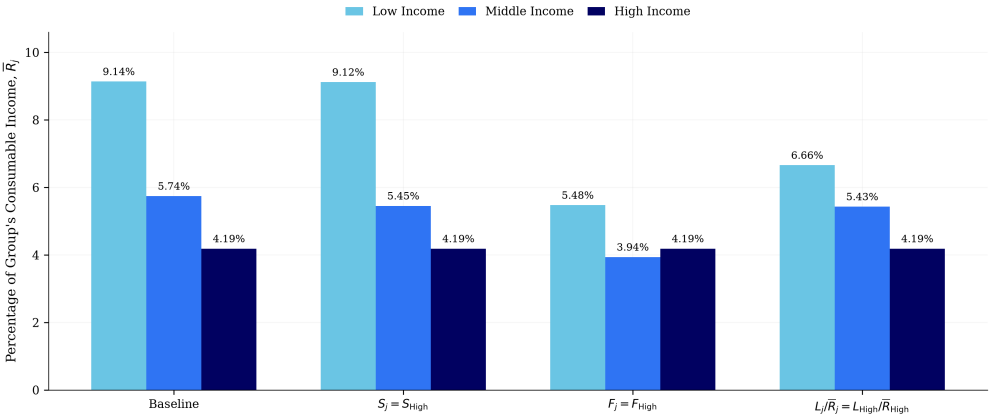
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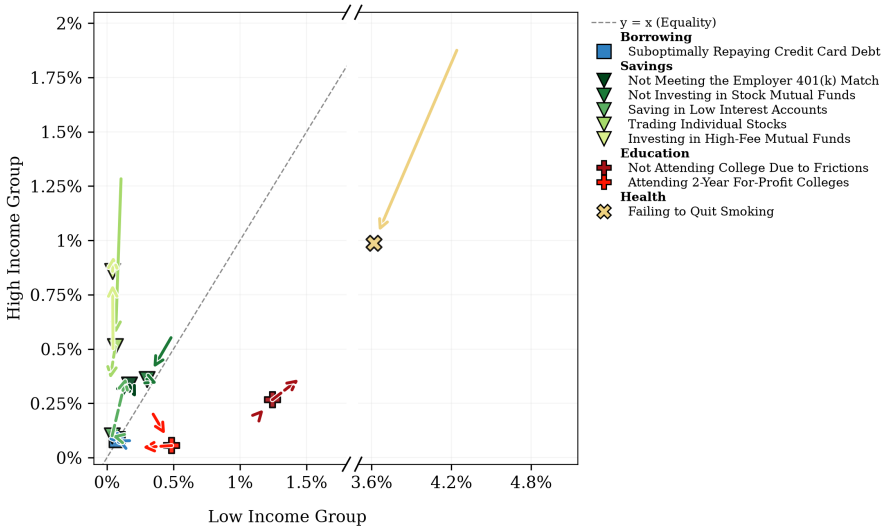
Decomposing Aggregate Financial Impacts



Decomposing Aggregate Financial Impacts



Evolution of Financial Impacts Over Time (In Progress)



(Points: 2010s; arrows: late 1990s/early 2000s → 2010s → 2020s)

Outline

1. Framework
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5. Conclusion

- How do our estimates of financial impact compare with **different sources?**

Coverage in Education, Research, and Policy

- How do our estimates of financial impact compare with **different sources?**
- **Education**: undergraduate financial literacy syllabi from 10 top universities
  - Columbia, Duke, Harvard, Princeton, Stanford, UC Berkeley, UC San Diego, Florida, UNC Chapel Hill, Yale
- **Education**: high school financial literacy state standards from 10 states
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Financial Impacts and Coverage Rankings

Source Top 3 Frictions	Count	Correlation of Rank in Source with Rank in $S_j F_j L_j / \overline{R}_j$ Low Income	 High Income

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Source Top 3 Frictions	Count	Correlation of Rank in Source with Rank in $S_j F_j L_j / \overline{R}_j$ Low Income	 High Income
Coverage in university syllabi Stock Market, Trading Individual Stocks, Refinancing	2.3	-0.367	0.250

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Coverage in high-school standards College, Stock Market, Late Fees	2.9	0.104	0.298

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Coverage in high-school standards College, Stock Market, Late Fees	2.9	0.104	0.298
Coverage in journal articles College, 401(k) Match, Stock Market	4.3	0.366	0.614

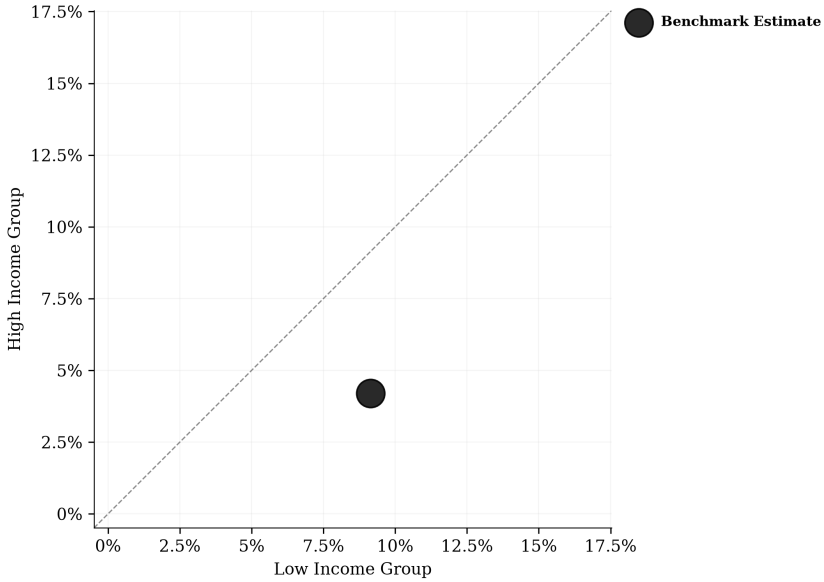
Financial Impacts and Coverage Rankings

Source Top 3 Frictions	Count	Correlation of Rank in Source with Rank in $S_j F_j L_j / \overline{R}_j$ Low Income	 High Income
Coverage in university syllabi Stock Market, Trading Individual Stocks, Refinancing	2.3	-0.367	0.250
Coverage in high-school standards College, Stock Market, Late Fees	2.9	0.104	0.298
Coverage in journal articles College, 401(k) Match, Stock Market	4.3	0.366	0.614
Coverage in Economic Report of the President: Democratic EITC, College, SNAP	3.7	0.412	0.130

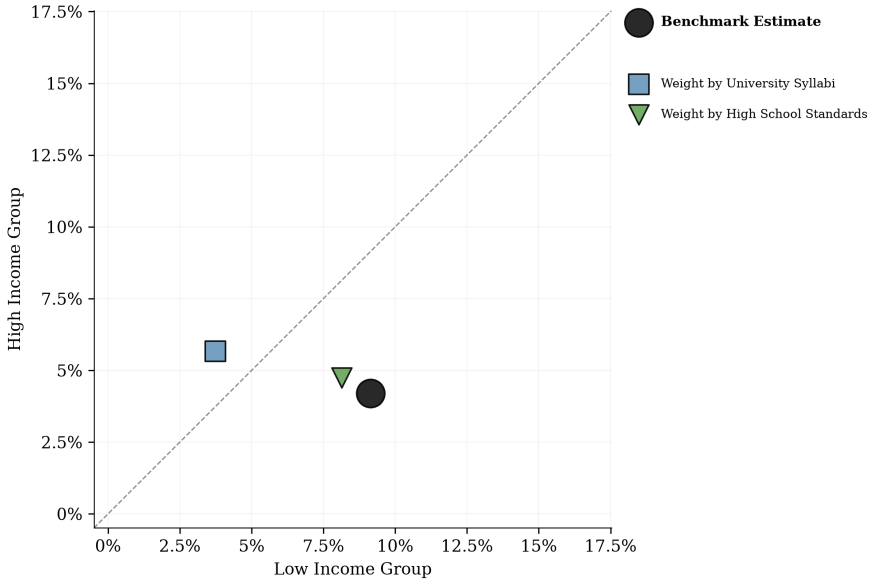
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Coverage in Economic Report of the President: Republican EITC, Low-Interest Accounts, SNAP	2.6	0.264	0.034

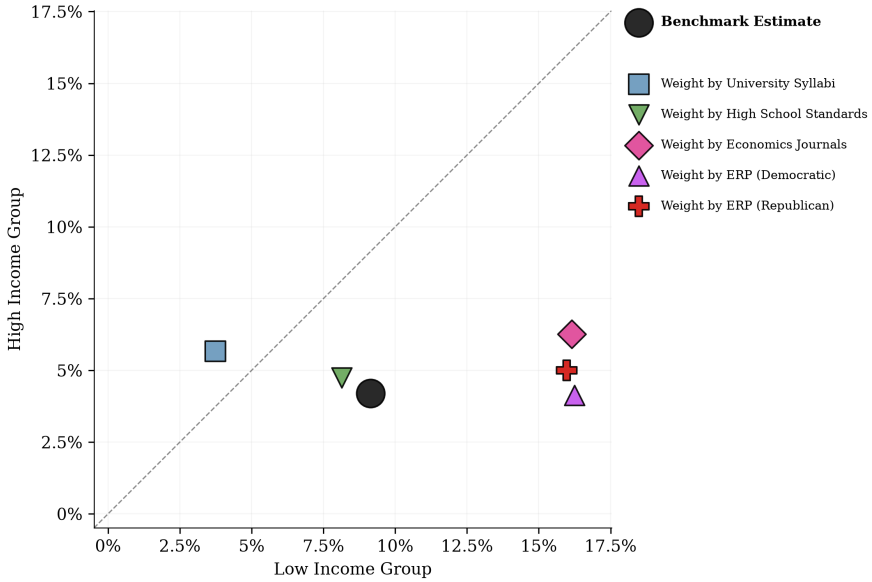
Coverage-Weighted Aggregate Financial Impacts



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Coverage-Weighted Aggregate Financial Impacts



Expert Forecasts of Financial Impacts

- Survey on the **Social Science Prediction Platform** (DellaVinga et al. 2019)
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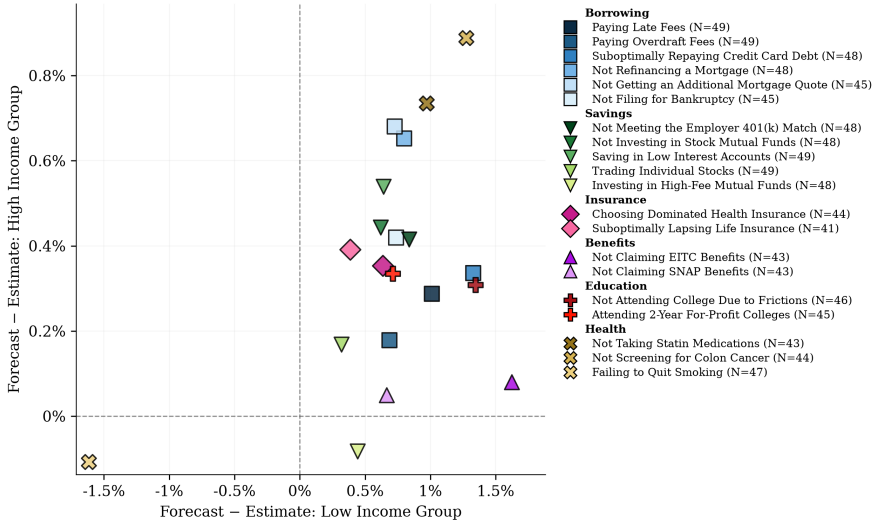
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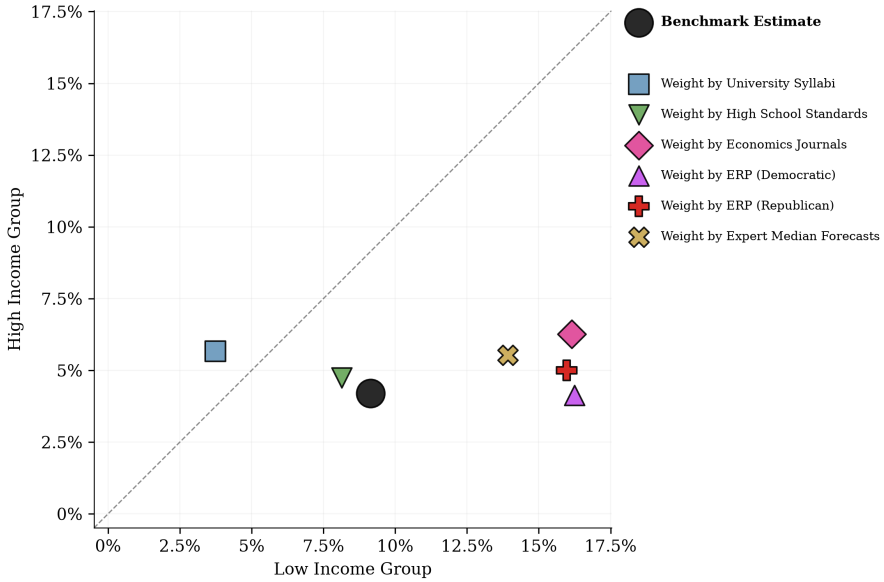
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Rankings of financial impacts from expert forecasts, high income Colon Cancer, Smoking, Statins	46.1	-0.174	0.665

Forecast Deviations = Forecast − Estimate



▸ Levels

Coverage-Weighted Aggregate Financial Impacts



Outline

1. Framework
2. Measurement and Examples
3. Summarizing Our Estimates
4. Comparison with Other Sources
5. Conclusion

Summary

1. Decision-making frictions create sizeable losses in aggregate
   - Aggregate losses are **9.1%** (**4.2%**) for low (high) income $\approx$ expenditure on utilities
   - Largest (as of now): smoking, fund fees, college attendance, claiming SNAP
   - **But** losses are typically **smaller** than experts estimate

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2. Losses from decision-making frictions are **regressive**
   - High-income have more to lose (L), but low-income **prevalence** (F) is higher
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   - Losses are **less regressive** than experts estimated

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... Much more to come!

Thank You!

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Companion Paper: A Life Cycle Model

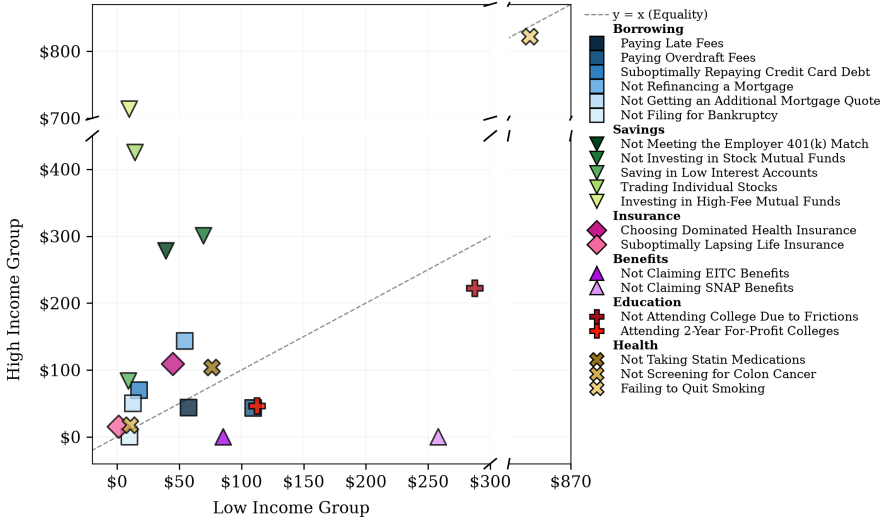
- Reasons to develop a life cycle model:
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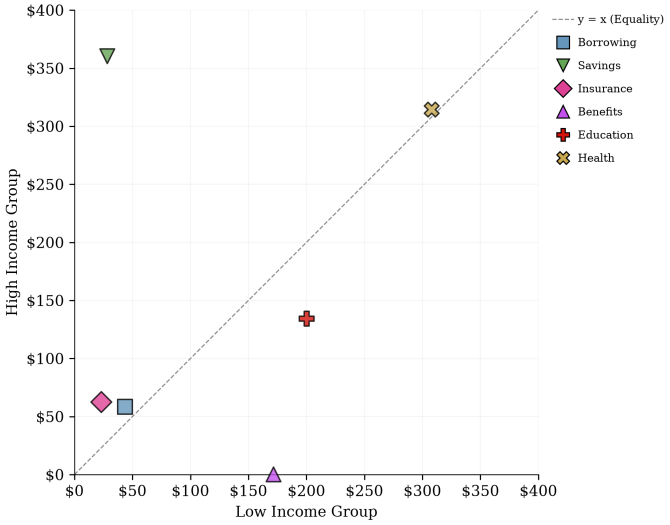
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- Individuals subject to frictions $\mathbf{f}_i \in \{0,1\}^{20}$ based on **empirical prevalence**
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- **Key counterfactuals**: set $\mathbf{f}_i(n) = 0$ for each friction n and compare with data

Financial Impacts Across Frictions: Dollars



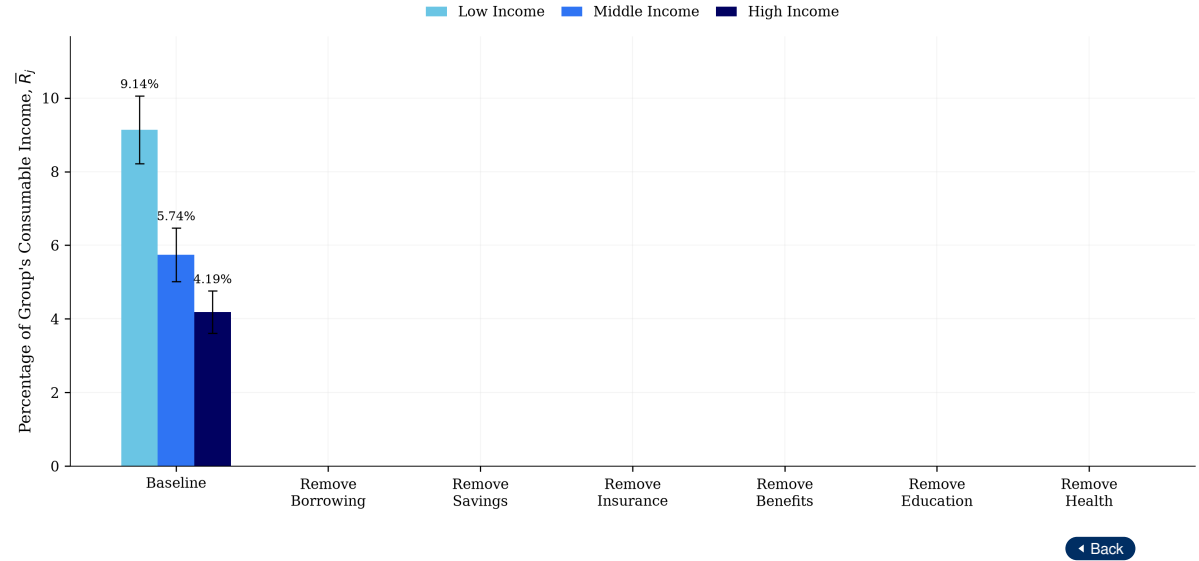
Back

Financial Impacts: Category Averages in Dollars



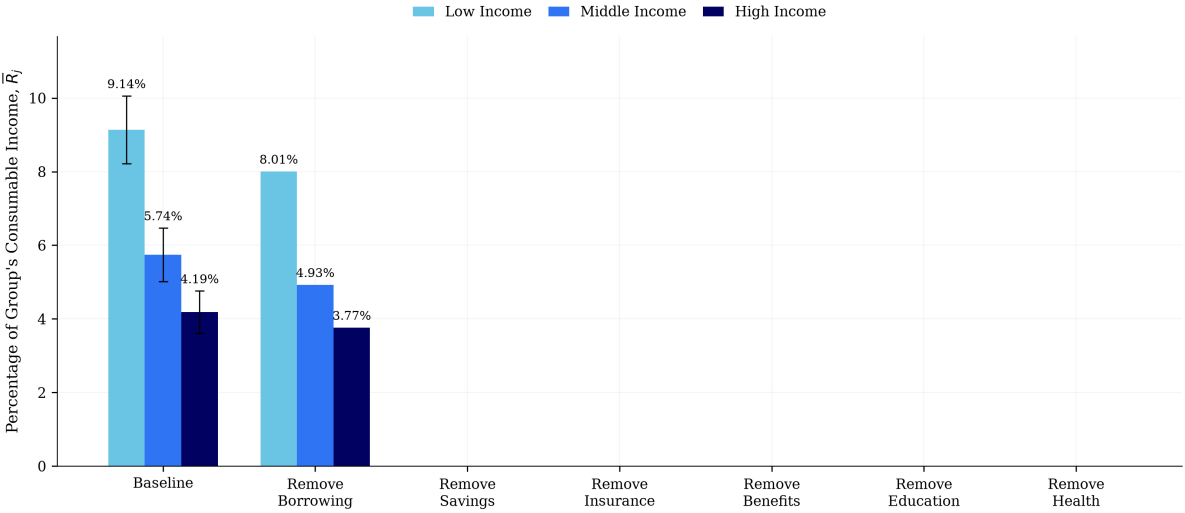
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Aggregate Financial Impact After Removing Each Category



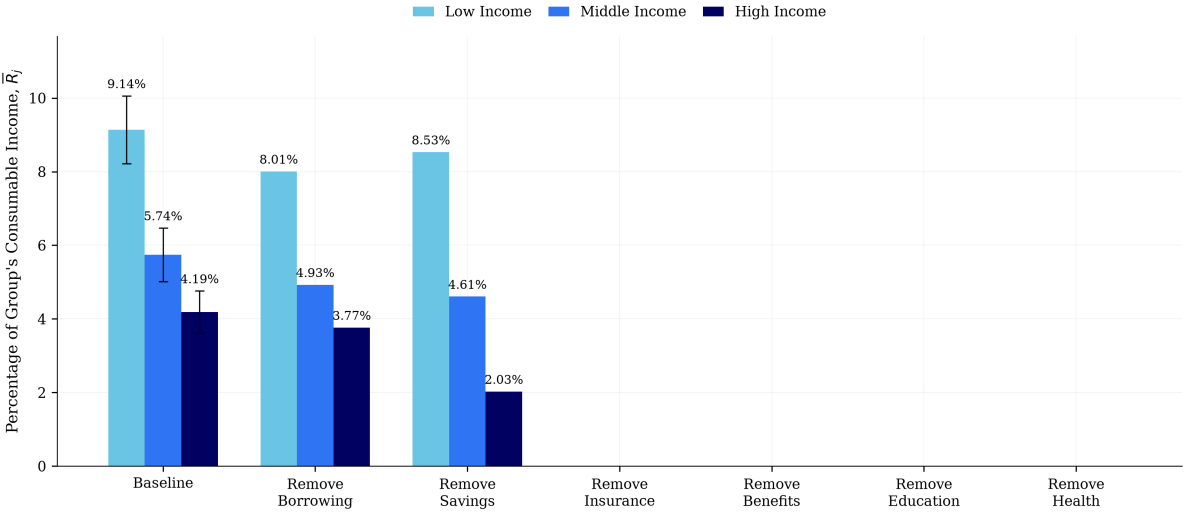
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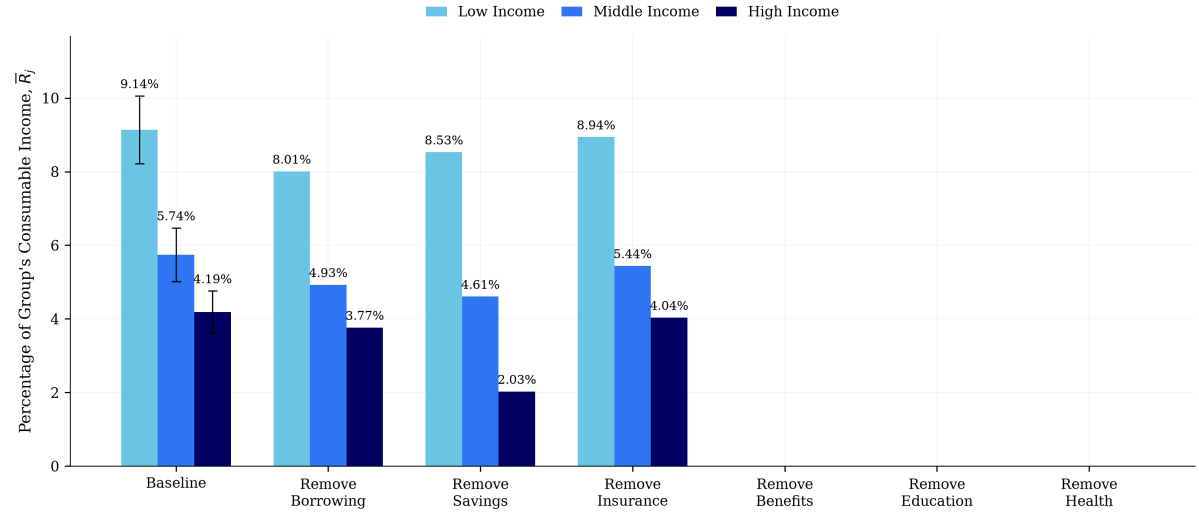
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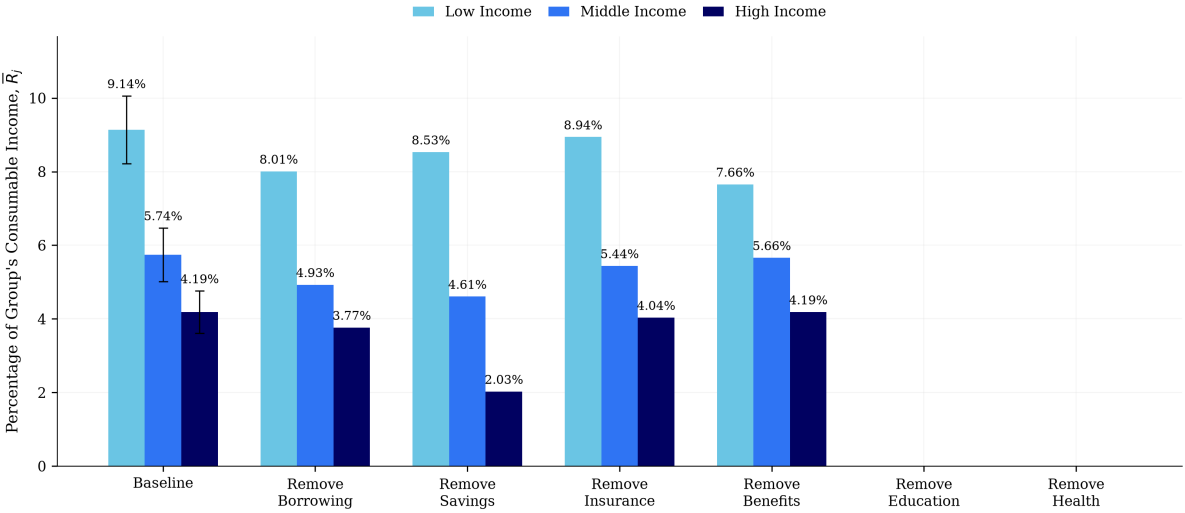
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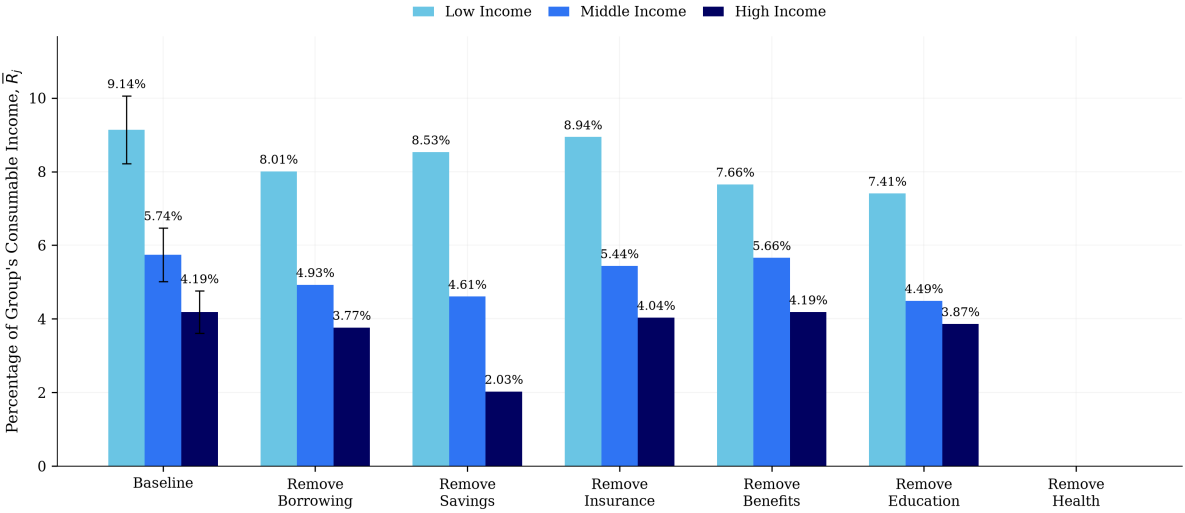
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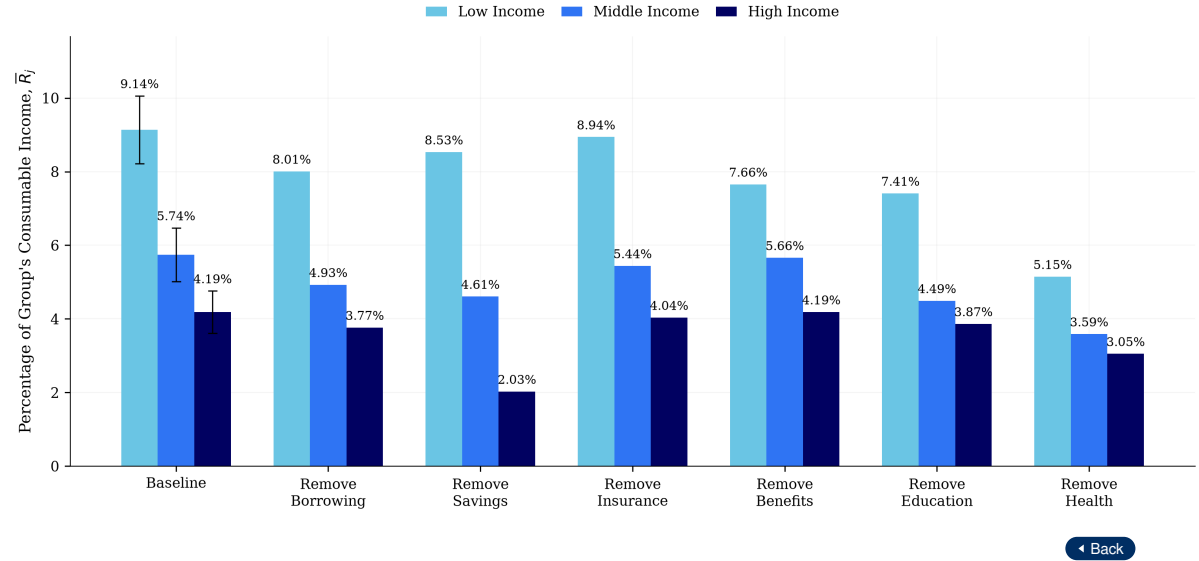
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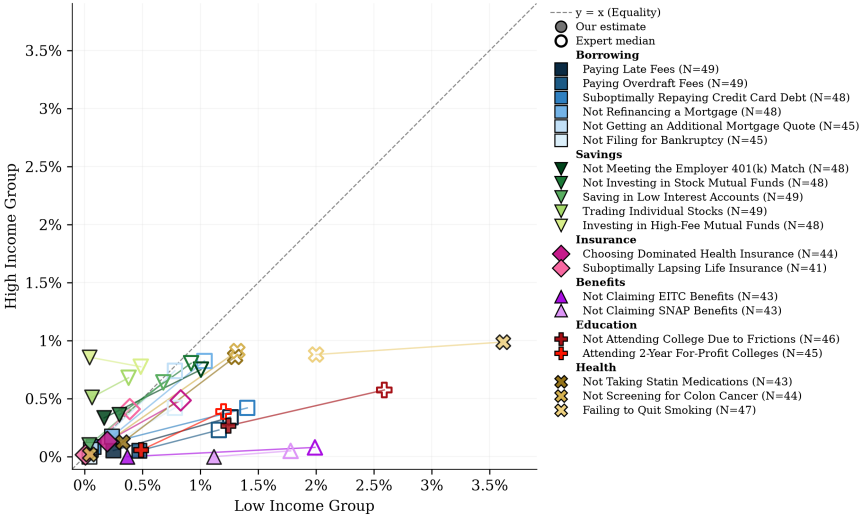
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Aggregate Financial Impact After Removing Each Category



Back

Expert Predictions and Our Estimates: Levels



Back

Example: Failing to Quit Smoking, 1997–2005

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.436	0.664	\$3,394	\$981
	(0.005)	(0.006)		
Middle Income	0.473	0.533	\$3,525	\$889
	(0.004)	(0.006)		
High Income	0.480	0.420	\$7,732	\$1,561
	(0.010)	(0.013)		

- Mortality ratio held at its 2010–2018 value
- Annualize losses from forgone consumable income and cigarette spending

$$L_{\text{Low}} = \overbrace{\$2{,}967}^{\text{mortality component}} + \overbrace{\$427}^{\text{cigarette spending}} = \$3{,}394 \text{ / year}$$

Back

List of Decision-Making Frictions

No.	Category	Friction	No.	Category	Friction
1	Borrowing	Paying Late Fees	11	Savings	Investing in High-Fee Mutual Funds
2	Borrowing	Paying Overdraft Fees	12	Insurance	Choosing Dominated Health Insurance
3	Borrowing	Suboptimally Repaying Credit Card Debt	13	Insurance	Suboptimally Lapsing Life Insurance
4	Borrowing	Not Refinancing a Mortgage	14	Benefits	Not Claiming EITC Benefits
5	Borrowing	Not Getting an Additional Mortgage Quote	15	Benefits	Not Claiming SNAP Benefits
6	Borrowing	Not Filing for Bankruptcy	16	Education	Not Attending College Due to Frictions
7	Savings	Not Meeting the Employer 401(k) Match	17	Education	Attending 2-Year For-Profit Colleges
8	Savings	Not Investing in Stock Mutual Funds	18	Health	Not Taking Statin Medications
9	Savings	Saving in Low Interest Accounts	19	Health	Not Screening for Colon Cancer
10	Savings	Trading Individual Stocks	20	Health	Failing to Quit Smoking

◂ Back: Late Fees ◂ Back: For-Profit ◂ Back: Stocks ◂ Back: Smoking ◂ Back: Impacts

Empirical Evidence of All Frictions (1/10)

Paying Late Fees

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.466	0.309	$409	$58
	(0.013)	(0.018)	[384, 429]	[49, 64]
Middle Income	0.738	0.248	$409	$73
	(0.004)	(0.010)	[384, 429]	[70, 78]
High Income	0.829	0.132	$409	$44
	(0.008)	(0.011)	[384, 429]	[38, 51]

Paying Overdraft Fees

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.777	0.432	$324	$110
	(0.015)	(0.020)	[293, 356]	[95, 127]
Middle Income	0.964	0.347	$300	$100
	(0.004)	(0.015)	[278, 328]	[90, 114]
High Income	0.992	0.172	$255	$43
	(0.003)	(0.012)	[215, 304]	[34, 55]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Suboptimally Repaying Credit Card Debt

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.143 (0.006)	0.882 (0.001)	\$140 [123, 156]	\$18 [15, 20]
Middle Income	0.301 (0.005)	0.882 (0.001)	\$192 [179, 210]	\$51 [47, 56]
High Income	0.279 (0.008)	0.882 (0.001)	\$283 [249, 320]	\$70 [60, 79]

Not Refinancing a Mortgage

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.157 (0.012)	0.833 (0.007)	\$348 [233, 487]	\$54 [35, 78]
Middle Income	0.383 (0.008)	0.702 (0.006)	\$188 [173, 213]	\$72 [66, 83]
High Income	0.499 (0.010)	0.672 (0.006)	\$288 [265, 335]	\$144 [132, 169]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Not Getting an Additional Mortgage Quote

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.150	0.601	\$140	\$13
	(0.011)	(0.000)	[106, 199]	[9, 19]
Middle Income	0.381	0.521	\$137	\$27
	(0.008)	(0.000)	[125, 148]	[25, 29]
High Income	0.495	0.474	\$213	\$50
	(0.009)	(0.000)	[196, 240]	[46, 57]

Not Filing for Bankruptcy

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.284	0.200	\$172	\$10
	(0.014)	(–)	[148, 208]	[8, 12]
Middle Income	0.131	0.200	\$130	\$3
	(0.007)	(–)	[117, 149]	[3, 4]
High Income	0.003	0.200	\$380	\$0
	(0.002)	(–)	[118, 622]	[0, 1]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Not Meeting the Employer 401(k) Match

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.184 (0.007)	0.568 (0.001)	\$373 [360, 385]	\$39 [36, 42]
Middle Income	0.488 (0.006)	0.299 (0.000)	\$1,013 [986, 1,040]	\$148 [143, 153]
High Income	0.646 (0.007)	0.200 (0.001)	\$2,154 [2,062, 2,249]	\$278 [266, 292]

Not Investing in Stock Mutual Funds

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.328 (0.009)	0.648 (0.013)	\$326 [277, 404]	\$69 [57, 87]
Middle Income	0.645 (0.006)	0.310 (0.008)	\$895 [803, 983]	\$179 [159, 200]
High Income	0.869 (0.006)	0.131 (0.007)	\$2,631 [2,116, 3,070]	\$301 [236, 357]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Saving in Low Interest Accounts

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.284	0.638	\$50	\$9
	(0.008)	(0.017)	[42, 61]	[7, 11]
Middle Income	0.455	0.602	\$102	\$28
	(0.006)	(0.009)	[91, 110]	[25, 30]
High Income	0.692	0.532	\$228	\$84
	(0.008)	(0.010)	[208, 251]	[77, 93]

Trading Individual Stocks

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.284	0.129	\$393	\$14
	(0.008)	(0.012)	[192, 625]	[6, 23]
Middle Income	0.455	0.212	\$290	\$28
	(0.006)	(0.008)	[239, 360]	[23, 35]
High Income	0.692	0.397	\$1,550	\$426
	(0.008)	(0.011)	[1,371, 1,779]	[376, 497]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Investing in High-Fee Mutual Funds

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.136 (0.006)	0.910 (0.029)	\$79 [64, 113]	\$10 [8, 15]
Middle Income	0.495 (0.007)	0.910 (0.029)	\$168 [161, 184]	\$76 [70, 84]
High Income	0.780 (0.008)	0.910 (0.029)	\$1,005 [936, 1,074]	\$714 [648, 779]

Choosing Dominated Health Insurance

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.064 (0.005)	0.620 (0.006)	\$1,121 [1,002, 1,243]	\$45 [37, 56]
Middle Income	0.200 (0.014)	0.520 (0.005)	\$1,121 [1,002, 1,243]	\$116 [96, 138]
High Income	0.244 (0.017)	0.400 (0.006)	\$1,121 [1,002, 1,243]	\$109 [89, 131]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Suboptimally Lapsing Life Insurance

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.121 (0.005)	0.180 (0.006)	\$49 [45, 61]	\$1 [1, 1]
Middle Income	0.284 (0.006)	0.171 (0.002)	\$113 [108, 117]	\$6 [5, 6]
High Income	0.338 (0.007)	0.158 (0.001)	\$293 [285, 309]	\$16 [15, 17]

Not Claiming EITC Benefits

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.384 (0.007)	0.201 (0.001)	\$1,100 [1,061, 1,149]	\$85 [81, 90]
Middle Income	0.060 (0.002)	0.196 (0.002)	\$1,003 [937, 1,080]	\$12 [11, 13]
High Income	0.001 (0.000)	0.222 (0.000)	\$949 [585, 1,439]	\$0 [0, 0]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Empirical Evidence of All Frictions (8/10)

Not Claiming SNAP Benefits

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.411 (0.007)	0.657 –	\$956 [927, 994]	\$258 [246, 271]
Middle Income	0.032 (0.002)	0.657 –	\$1,030 [903, 1,166]	\$22 [19, 25]
High Income	0.001 (0.001)	0.657 –	\$192 [0, 192]	\$0 [0, 0]

Not Attending College Due to Frictions

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.497 (0.025)	0.403 (0.075)	\$1,438 [686, 2,159]	\$288 [131, 496]
Middle Income	0.351 (0.012)	0.280 (0.070)	\$3,967 [2,087, 5,768]	\$390 [163, 690]
High Income	0.196 (0.015)	0.129 (0.023)	\$8,793 [4,746, 12,671]	\$222 [107, 357]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Attending 2-Year For-Profit Colleges

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.263 (0.017)	0.236 (0.015)	$1,813 [1,522, 2,107]	$113 [85, 141]
Middle Income	0.277 (0.009)	0.094 (0.007)	$4,320 [3,593, 5,065]	$113 [85, 139]
High Income	0.138 (0.011)	0.037 (0.013)	$9,165 [7,619, 10,783]	$46 [14, 78]

Not Taking Statin Medications

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.255 (0.005)	0.469 (0.024)	$638 [497, 800]	$76 [59, 95]
Middle Income	0.248 (0.004)	0.404 (0.023)	$723 [565, 908]	$72 [57, 91]
High Income	0.241 (0.008)	0.357 (0.019)	$1,211 [943, 1,491]	$104 [81, 128]

◂ Back: Frictions ◂ Back: Sources ◂ Index

Not Screening for Colon Cancer

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.484	0.621	$35	$11
	–	(0.005)	[23, 46]	[7, 14]
Middle Income	0.532	0.526	$41	$11
	–	(0.004)	[30, 52]	[8, 14]
High Income	0.561	0.446	$70	$18
	–	(0.006)	[53, 87]	[13, 22]

Failing to Quit Smoking

Income Group (j)	S_j	F_j	L_j	$S_j F_j L_j$
Low Income	0.437	0.624	$3,066	$837
	(0.005)	(0.008)	[2,920, 3,516]	[797, 960]
Middle Income	0.452	0.465	$3,718	$781
	(0.004)	(0.006)	[3,439, 4,317]	[723, 907]
High Income	0.396	0.352	$5,893	$822
	(0.006)	(0.011)	[4,825, 11,680]	[673, 1,630]

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